



Common and country specific economic uncertainty[☆]



Haroon Mumtaz^{a,*}, Konstantinos Theodoridis^b

^aQueen Mary College, United Kingdom

^bBank of England, United Kingdom

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ABSTRACT

We use a factor model with stochastic volatility to decompose the time-varying variance of macroeconomic and financial variables into contributions from country-specific uncertainty and uncertainty common to all countries. We find that the common component plays an important role in driving the time-varying volatility of nominal and financial variables. The cross-country co-movement in volatility of real and financial variables has increased over time with the common component becoming more important over the last decade. Simulations from a two-country DSGE model featuring Epstein-Zin preferences suggest that increased globalisation and trade openness may be the driving force behind the increased cross-country correlation in volatility.

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1. Introduction

Recent turmoil in financial markets has led to a substantial increase in macroeconomic volatility across the industrialised world. This is clear from the simple calculation in Fig. 1 which shows the average of the rolling standard deviation of the main macroeconomic and financial variables for eleven OECD countries. This simple measure of economic volatility shows an increase across all countries over the post-2007 period highlighting the severity of the financial crisis. It is interesting to note that this high correlation of volatility is not just confined to the recent financial crisis but appears to be a prominent feature of this statistic over several episodes in the past. A casual examination of the figure suggests that this measure of volatility moved especially closely together during the mid-1970s, the early 1980s and then during the beginning and end of the last decade. The full sample correlation between these volatility measures is high, averaging across pairwise combinations at 50%.

The aim of this paper is to investigate the co-movement in volatility from empirical and theoretical perspectives. Using a dynamic factor model with stochastic volatility, we decompose the movements in the volatility of real activity, inflation and financial series from these eleven OECD countries into the contributions from country-specific and OECD wide uncertainty. We find that OECD wide uncertainty plays an important role in driving the variance of real and nominal variables and is especially important for the latter series. Moreover, we estimate that the contribution of common uncertainty has increased over the sample period and the volatility of key variables displays a higher correlation after the late 1990s.

We then build a two country model that features households with Epstein-Zin preferences. One of the key implications of the model set-up is the presence of heteroscedastic endogenous variables, with this feature induced by agents' preferences. We show that in a two country environment, shocks that lead to transfer of resources across countries imply co-movement in the second moments of endogenous variables. This co-movement rises as trade links in the model become stronger. In contrast, changes in monetary policy rule and/or the Phillips curve are unable to explain the increase in the importance of the common component in volatility.

This paper is closely related to two recent strands of the empirical Macro literature. The aim of the paper is similar in spirit to the work on international business cycles (see for example Kose et al., 2003)

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* Corresponding author.

E-mail addresses: h.mumtaz@qmul.ac.uk (H. Mumtaz), Konstantinos.Theodoridis@bankofengland.co.uk (K. Theodoridis).

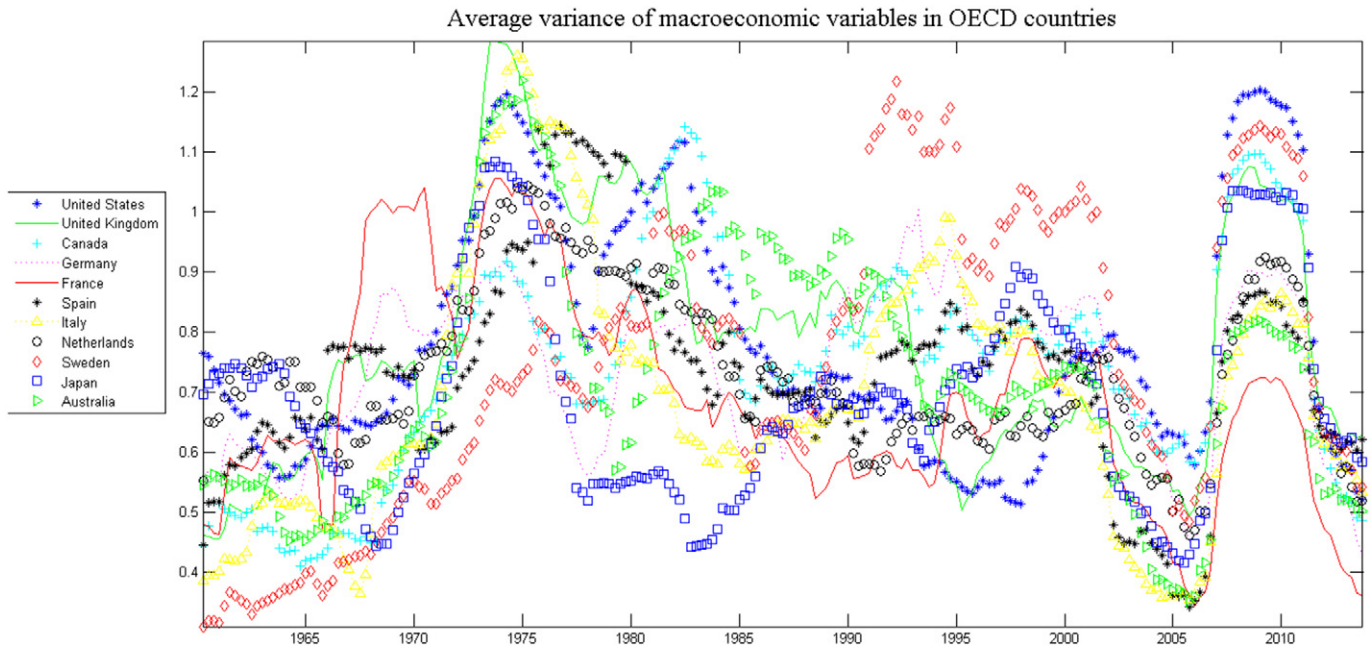


Fig. 1. Two year rolling estimate of the average variance of macroeconomic variables in 11 OECD countries. The data set for each country is described in Section 2.2. For the purposes of this figure, the rolling standard deviation is calculated for each series and then the mean across the series is calculated. The data is standardised prior to this calculation.

and the research on inflation co-movements (see Mumtaz and Surico, 2008) that has sought to establish the importance of a common factor in explaining the movements in these variables. We focus on co-movement in the second moment and show that this feature is important from empirical and theoretical perspectives. Our analysis is also closely related to the recent literature on uncertainty that has focussed on estimating proxies for economic uncertainty. However, we investigate the role of *common* movements in uncertainty, an aspect that has been relatively unexplored in this literature. This investigation is related to the analysis by Berger et al. (2014) and Berger et al. (2016) who estimate global uncertainty associated with output and inflation. However, our work explores common *macroeconomic* uncertainty that encompasses both macroeconomic and financial variables. Moreover, in contrast to Berger et al. (2014) and Berger et al. (2016), a major aim of the current paper is to build a DSGE model to provide a theoretical explanation for the empirical findings.

Our results have important policy implications. In particular, they highlight potential consequences of globalisation that have been largely ignored in the literature – i.e. an increasing co-movement in the time-varying volatility of output, inflation and stock returns. This suggests that policy makers may need to focus more on economic developments that cause uncertainty in countries that are linked through trade. This implication is especially relevant given the recent debt crisis in the Euro Area that has resulted in an increase in volatility in a number of countries.

The paper is organised as follows: Sections 2 and 2.1 introduce the empirical model and discuss the estimation method. The results from the empirical model are presented in Section 3. We introduce the DSGE model and present the model simulations in Section 4.

2. Common and country-specific uncertainty

In order to estimate country-specific and common (or ‘world’) specific measures of uncertainty, we use a dynamic factor model with time-varying volatility. The factor model is defined as

$$X_{it} = B_t^C F_t^C + B_t^W F_t^W + e_{it} \quad (1)$$

where X_{it} is a panel of macroeconomic and financial data for the set of OECD countries described below. This panel of data is summarised by three components: a set of K_W common or ‘world’ factors F_t^W , a set of K_C country-specific factors F_t^C and idiosyncratic components e_{it} . The world and the country factors follow VAR processes:

$$F_t^W = \zeta + \sum_{j=1}^P p_j F_{t-j}^W + \Upsilon_t^{1/2} g_t \quad (2)$$

$$F_t^C = \alpha + \sum_{j=1}^P p_j F_{t-j}^C + \Omega_t^{1/2} v_t \quad (3)$$

while the idiosyncratic components have an AR transition equation

$$e_{it} = \sum_{j=1}^J \mu_{ij} e_{it-j} + h_{it}^{1/2} \varepsilon_{it} \quad (4)$$

where $g_t, v_t, \varepsilon_{it} \sim N(0,1)$. Note that the error terms in Eqs. (2), (3) and (4) are heteroscedastic. The error covariance matrices in the VAR models (2) and (3) are defined as

$$\Upsilon_t = C^{-1} D_t C^{-1'} \quad (5)$$

$$\Omega_t = A^{-1} H_t A^{-1'} \quad (6)$$

where A and C are lower triangular and D_t and H_t are diagonal matrices defined as

$$D_t = \text{diag}(s_k \gamma_t) \quad (7)$$

$$H_t = \text{diag}(S_k \lambda_t).$$

The time-varying volatility is captured by γ_t and λ_t with s_k and S_k the scaling factors for $k = 1, 2, \dots, K_m$ for $m = W, C$. The overall volatilities evolve as an AR(1) process:

$$\ln \gamma_t = \bar{\alpha} + \bar{\beta} \ln \gamma_{t-1} + \bar{Q}^{1/2} \bar{\eta}_t \quad (8)$$

Table 1
Model comparison via DIC. Best fit indicated by lowest DIC.

	DIC
2 factors	1.65×10^5
3 factors	1.75×10^5
4 factors	1.52×10^5
5 factors	1.64×10^5

$$\ln \lambda_t = \tilde{\alpha} + \tilde{\beta} \ln \lambda_{t-1} + \tilde{Q}^{1/2} \tilde{\eta}_t. \quad (9)$$

The structure defined by Eqs. (5) and (6) suggests that the volatility specification is characterised by the following features: First, the specification captures the *overall* volatility in the orthogonalized residuals of the VAR models. As explained by [Carriero et al. \(2012\)](#), the common volatilities can be interpreted as the average of the variance of the shocks with equal weight given to each individual volatility. Note that the errors to these equations represent the shocks to ‘world’ and country factors. Thus, γ_t and λ_t capture the average volatility of the unpredictable part of the common component and the country-specific component. We interpret these volatilities as measures of uncertainty associated with OECD wide economic conditions and country-specific economic conditions.

The variance of the shocks to the idiosyncratic components are also assumed to be heteroscedastic with h_{it} evolving as a stochastic volatility process

$$\ln h_{it} = a_i + b_i \ln h_{it-1} + q_i^{1/2} n_{it}. \quad (10)$$

The structure of the model implies that the variance of each series at time t can be written as a function of γ_t , Ω_t and h_t . In particular

$$\text{var}(X_{it}) = (B_i^C)^2 \text{var}(F_t^C) + (B_i^W)^2 \text{var}(F_t^W) + \text{var}(e_{it}) \quad (11)$$

where the variance terms on the RHS of Eq. (11) can be calculated using the standard VAR formula for the variance of the endogenous variables. Note that these variance terms are time-varying as they are functions of λ_t , γ_t and h_{it} respectively. The volatility of each series in our panel is thus driven by uncertainty that is common to all countries, uncertainty that is country-specific and a residual term that captures sectoral volatility and data uncertainty. Our framework, therefore, allows us to calculate how volatility of key series (such as GDP growth, CPI inflation and stock market returns) is driven by uncertainty that is common to all countries and uncertainty that is country- and series-specific.

This empirical model is related to a number of contributions in the recent literature on measuring uncertainty. In particular, the spirit of our model is similar to the procedure used by [Jurado et al. \(2013\)](#) to estimate US economic uncertainty. The uncertainty measure by [Jurado et al. \(2013\)](#) is the average time-varying variance in the unpredictable component of a large set of real and financial time-series. The volatility specification in our factor model has a similar interpretation – it attempts to capture the average volatility in the shocks to the factors that summarise real and financial conditions. In contrast to [Jurado et al. \(2013\)](#), however, our model allows the estimation of uncertainty at the country and at the ‘world’ level.¹

It is interesting to note that this specification is more general than those employed by [Mumtaz and Surico \(2008\)](#), [Berger et al. \(2014\)](#) and [Berger et al. \(2016\)](#). These studies focus on the volatility of the unpredictable component of output growth or inflation factors. In particular, these studies estimate factor models of the following *basic* form:

$$y_{it} = \Gamma_{R,i} R_t + \Gamma_{\tilde{C},i} \tilde{C}_t + \varepsilon_{it} \quad (12)$$

where y_{it} is chosen to be a cross-country panel of inflation rates by [Mumtaz and Surico \(2008\)](#), and GDP growth by [Berger et al. \(2016\)](#).² In this model, R_t represents an inflation or GDP growth factor that is common across countries with Γ_i denoting the factor loadings. Similarly, $\tilde{C}_t$ denotes the country-specific factor in either inflation or GDP growth with the associated factor loadings given by $\Gamma_{\tilde{C}}$. The dynamics of R_t and $\tilde{C}_t$ are assumed to be described by uni-variate AR processes that feature stochastic volatility. These stochastic volatilities are interpreted as uncertainty at the ‘world’ and country levels. These models differ from our specification in two key dimensions: First, unlike [Mumtaz and Surico \(2008\)](#) and [Berger et al. \(2016\)](#), our data set features a range of macroeconomic and financial variables for each country included in the panel. This means that the set of factors F_t^W and F_t^C in Eq. (1) capture common and country-specific *economic* conditions and not simply common inflation or GDP growth. Second, the law of motion for the set of factors F_t^W and F_t^C are assumed to be VARs rather than uni-variate AR processes. The variance of the VAR shocks is modelled using the common stochastic volatility specification of [Carriero et al. \(2012\)](#). This common volatility summarises the uncertainty in common and country-specific economic conditions as captured by F_t^W and F_t^C . We are thus able to interpret the resulting uncertainty measures as pertaining to *economic* uncertainty rather than a narrower measure focussing on a particular variable.

2.1. Estimation and model specification

The factor model in Eqs. (1) to (10) is estimated via Gibbs sampling. The technical Appendix provides details of the priors and the conditional posterior distributions. In the benchmark specifications, we use 50,000 replications and base our inference on the last 5000 replications. The recursive means of the retained draws (see technical Appendix) show little fluctuation providing support for convergence of the algorithm.³

In order to maintain parsimony, P , the lag lengths in the VARs are fixed at 2. In addition, we allow for first order serial correlation in the idiosyncratic errors e_{it} . The number of common and country-specific factors is an important specification choice. For the model selection exercise, we set $K_W = K_C$ in Eq. (1). We consider models with 2 to 5 common and country factors and select the model which minimises the Bayesian Deviance Information Criterion (DIC).⁴

Introduced by [Spiegelhalter et al. \(2002\)](#), the DIC is a generalisation of the Akaike information criterion – it penalises model complexity while rewarding fit to the data. As shown in the technical Appendix, the DIC can be calculated as $DIC = \bar{D} + p_D$ where $\bar{D}$ measures goodness of fit and p_D approximates model complexity. A model with a lower DIC is preferred. Table 1 shows that the DIC is

² Note that we ([Berger et al., 2016](#)) include each country's inflation as an exogenous regressor.

³ The technical Appendix presents results from a small Monte-Carlo experiment that shows that this MCMC algorithm performs well.

⁴ We vary the country and common factors in a symmetric manner in this exercise. In theory, it is possible to consider combinations of a different number of these factors. Note, however, that as a computationally intensive particle filter algorithm (see Appendix) is used to compute the likelihood for the purpose of DIC calculation, computational feasibility limits us to the symmetric approach.

¹ [Mumtaz and Theodoridis \(2014\)](#) use a factor model with common stochastic volatility to estimate the time-varying impact of uncertainty shocks on the US economy. Their specification does not, however, distinguish between OECD and country-specific uncertainty.

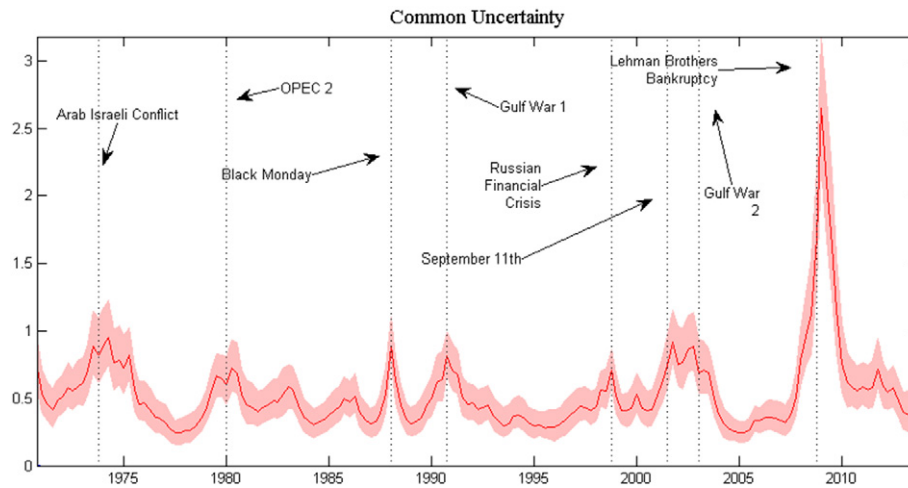


Fig. 2. The estimate of the common standard deviation of shocks to the world factors.

minimised for the model with 4 world and country factors and thus we set $K = 4$ in our benchmark model.

2.2. Data

As alluded to above, the model is estimated using quarterly data on eleven OECD countries. We consider data for the United States, United Kingdom, Canada, Germany, France, Spain, Italy, The Netherlands, Sweden, Japan and Australia. We limit our attention to these industrial countries mainly because of data availability – data on real and financial sectors of the economy is available for a reasonably long time span for these countries. Both of these features are essential for our analysis as we aim to incorporate uncertainty in a broad range of series and to also include periods which were subject to interesting fluctuations (such as the 1970s).

For each country, the data runs from 1960Q1 to 2013Q3. The number of series included for each country varies according to data availability. However, we attempt to maintain a similar composition of macroeconomic and financial series. For each country, the data set includes real activity variables (for e.g. exports, imports, consumption, investment, production, GDP), measures of inflation and earnings, interest rates and term spreads, corporate bond spreads, exchange rates and stock prices. Where relevant and available, oil prices are also included in the country-specific data sets. Where necessary, the variables are log differenced to induce stationarity. Finally, all series are standardised. The technical Appendix provides a list of the series used and the data sources.

3. Empirical results

3.1. The estimate of common and country-specific uncertainty

Fig. 2 presents the posterior estimate of $\gamma_t^{1/2}$, the common standard deviation of the shocks to the ‘world’ factors. We interpret this as a measure of uncertainty that is common across our panel of countries. It is interesting to note that this measure of common or ‘world uncertainty’ was at its highest over the recent financial crisis in 2009, again emphasizing the scale of the recent recession and its wide-ranging effect on the countries in our sample. The vertical lines in the figure indicate that previous peaks in this measure have coincided with important events such as wars in the Middle East, oil shocks, major financial crises and the September 11th terrorist attacks. This measure of common uncertainty shows noticeable differences from the output growth based measure presented

by Berger et al. (2016). In particular, common output growth uncertainty shown in Berger et al. (2016) remains relatively muted during periods of financial stress and political uncertainty during the early and late 1980s and at the beginning of the 2000s. In contrast, these events can be clearly linked to peaks in our measure as we account for financial stress by including financial variables and capture major global political events through their impact on asset prices such as oil and exchange rates.⁵

Fig. 3 shows the estimated country-specific uncertainty measures – i.e. $\lambda_{t,c}^{1/2}$ for countries $c = 1, 2, \dots, 11$. The estimated country-specific uncertainty for the United States reached its peak during the early 1980s that were characterised by a recession and change in the practice of monetary policy referred to as the ‘Volcker Experiment’. The mid-1980s saw another increase in US-specific uncertainty corresponding to the savings and loan crisis in 1985–1986. The late 1980s and early 1990s were also characterised by periods of increased uncertainty with the US economy buffeted by a stock market crash, recession and the onset of a credit crunch and the war in Iraq. Uncertainty declined after the mid-1990s but rose following the September 11th terrorist attacks in 2001 remaining elevated until the middle of the decade. Uncertainty was also higher during the recent recession. Note that this estimate of US-specific uncertainty has a correlation of 0.5 with the measure proposed by Jurado et al. (2013). While the two measures coincide during some time periods, several peaks in the Jurado et al. (2013) uncertainty measure are classified as common or ‘world’ uncertainty by our model.⁶

The UK-specific uncertainty measure peaked during the Sterling crisis of 1976. In the following years, this measure displays spikes during the recession of early 1980s, the exchange crisis in 1985, and the ERM exit in 1992. The inflation targeting period was characterised by low volatility until the recent recession when the measure increased again. Canadian uncertainty peaked during 1981, when the country experienced one of its worst recessions. Prominent increases in volatility can also be seen during the early 2000s when the Canadian economy experienced a slowdown coinciding with a large appreciation of the Canadian dollar. Finally, Canada-specific uncertainty increased sharply during the recent recession. German

⁵ To verify this, we re-estimate our empirical model eliminating any non-real activity series from our panel. The resulting measure of common uncertainty (see technical Appendix) is similar to the measure of Berger et al. (2016), but appears disconnected with periods of financial stress.

⁶ The correlation between the Jurado et al. (2013) measure and the sum of US-specific and common uncertainty is 0.6.

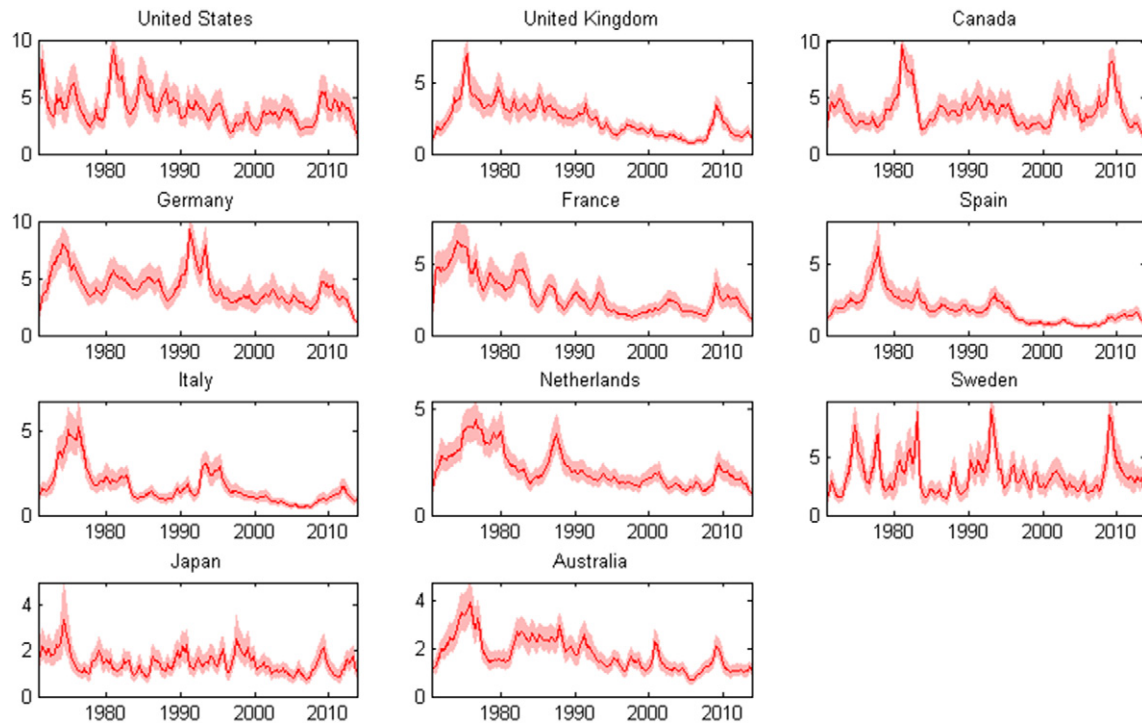


Fig. 3. The estimate of the common standard deviation of shocks to the country factors.

uncertainty is at its highest during the period of re-unification and the ERM crisis in the early 1990s. Note also that the two strongest post-war German recessions in 1974 and 2008 are characterised by increases in uncertainty with the former rise estimated to be substantially larger and the latter episode largely captured by the rise in world-specific uncertainty. The uncertainty in the remaining Euro-Area countries follows a similar pattern, with highs estimated during the oil shock of the early 1970s, the stock market crash of the late 1980s, the recessions following the ERM crisis and then during the great recession.⁷ It is interesting to note that uncertainty in Sweden appears to fluctuate more than the Euro-Area countries, with large increases coinciding with the oil shocks of the 1970s, the large depreciation in the exchange rate in 1982, the financial crisis of the early 1990s and finally during the downturn in 2008–2009.

Japanese uncertainty was high during the mid-1970s. It rose again with the bursting of the real-estate bubble in 1990–1991, in the aftermath of the Asian financial crisis and then in 2008/2009. In Australia, country-specific uncertainty was at its peak during the recession of the mid-1970s. Uncertainty remained elevated through the 1980s peaking after the stock market crash of 1987. Uncertainty rose again in 1991, when the Australian economy was embroiled in a deep recession. The slowdowns in 2001 and then in 2008/2009 also coincided with increasing uncertainty.

3.2. Variance decomposition

One of the key questions that we wish to address using our empirical model is to investigate how important the common uncertainty measure has been in driving the volatility of key macroeconomic and financial variables. We do this by decomposing the unconditional variance into contributions from λ_t , γ_t and h_{it} respectively. The

structure of the model implies that these contributions can be calculated simply using Eq. (11). Note that as the variance in the model is time-varying, the implied decomposition changes over time as well.

Table 2 presents the contribution of ‘world’ or common, country-specific and idiosyncratic uncertainties to the unconditional volatility of output growth, inflation, the short-term government rate and stock returns. Note that the table presents the average of the contributions across time. The first three columns of the table show the decomposition for output growth. Output growth is defined as the quarterly growth of GDP for all countries except Sweden where this variable is unavailable and we report the results for the volatility of the growth of industrial production. The country-specific uncertainty is, in general, estimated to be more important for output volatility than ‘world’ uncertainty implying that domestic uncertainty shocks are largely responsible for driving this variable. However, the role played by common uncertainty is non-negligible. On average, this contribution ranges from a modest 2% to 5% for the United States and the United Kingdom to a high of around 35% for France. The average contribution is greater than 10% for countries other than the United States and United Kingdom.

Columns 4 to 6 of Table 2 decompose the unconditional volatility of CPI inflation. Across countries, common uncertainty is estimated to be more important for inflation volatility than the variance of output growth, with a cross-country mean contribution of 26%. Average contributions are around 40% to 50% for countries such as the United States, Canada, Italy and Spain, with the contribution in Japan, Australia, The Netherlands and Sweden ranging between 14% to 30%. In contrast, the importance of ‘world’ uncertainty is estimated to be small for inflation variance in the United Kingdom, Germany and France. For these countries, inflation variance is persistently high during the 1970s and the early 1980s with this pattern shared by country-specific uncertainty for the United Kingdom and Germany and by idiosyncratic uncertainty for France.

Columns 7 to 9 of the table show the decomposition of the unconditional variance of the short-term government rate. Note that this variable was not included in the data set for Japan and Spain

⁷ Note that we do not include CDS spreads or ratings data for Euro-Area countries as these series are not available over the sample used to estimate the model. This may potentially affect the estimates of country-specific uncertainty towards the end of the sample.

Table 2

Contribution of the common, country-specific and idiosyncratic components to the variance of output growth, inflation, the short-term interest rate and stock returns.

	Output growth			Inflation			Interest rate			Stock ret.		
	Common	Country	Idio.	Common	Country	Idio.	Common	Country	Idio.	Common	Country	Idio.
United States	1.6	94.9	3.1	50.7	28.8	15.9	13.4	40.4	39.5	23.5	55.2	17.4
United Kingdom	4.2	88.6	6.1	2.4	97.3	0.2	1.6	20.5	75.9	2.7	96.8	0.4
Canada	16.7	14.8	65.0	39.4	44.7	11.6	9.7	7.5	80.2	43.5	2.9	52.0
Germany	21.0	22.9	50.2	6.6	92.5	0.6	37.0	0.9	60.7	51.9	10.7	32.2
France	35.8	4.7	55.1	4.6	20.5	71.0	13.5	0.3	85.7	42.7	23.5	29.0
Spain	23.1	1.6	72.8	43.1	56.1	0.4				27.0	26.5	41.2
Italy	27.3	41.5	26.8	58.6	41.1	0.1	14.6	6.3	77.2	46.5	6.3	45.1
The Netherlands	12.7	64.0	20.0	28.6	71.2	0.2	15.2	28.9	50.8	72.8	2.0	23.8
Sweden	10.5	22.5	62.1	19.2	80.6	0.2	8.6	2.0	86.3	44.0	24.9	24.6
Japan	18.1	49.3	26.7	14.0	84.8	1.0				14.7	84.7	0.4
Australia	9.7	89.5	0.6	19.9	80.0	0.0	31.1	0.5	68.0	26.2	73.2	0.3
Average	16.4	44.9	35.3	26.1	63.4	9.2	16.1	11.9	69.4	35.9	37.0	24.2

because the available data over some sub-samples was characterised by discrete changes in this variable, with this feature resulting in estimation difficulties.⁸ For countries such as the United States, Germany, The Netherlands and Australia, the contribution of common uncertainty to the volatility of this variable is higher than the estimate for output growth variance. It is interesting to note that idiosyncratic uncertainty plays a major role. If one interprets the idiosyncratic component of the interest rate as an approximation to the monetary policy shock, then this suggests that monetary policy uncertainty has played an important role in driving the variance of short-term rates.

'World' uncertainty also makes an important contribution to the variance of stock returns (columns 10 to 12 of Table 2). For example, the average contribution of common uncertainty is around 25% for the United States and Australia and ranges from 25% to over 70% for Canada and the Euro-Area countries. On average, common uncertainty is estimated to be more important for stock return variance than the remaining variables considered above.

3.2.1. Changes in the contributions over time

In Tables 3 and 4, we consider the temporal evolution of the contribution of the common, country-specific and idiosyncratic uncertainties to the variance of output growth, the short-term government rate and stock returns. The tables show the distribution of the percentage difference in the average contribution calculated after and before 1989Q4, with positive numbers indicating that the post-1990 contribution was larger.

Consider the change in the contribution of the common component. It is clear from the last row of the table that, on average, the post-1990 contribution of common uncertainty was higher for the variance of all variables, with the largest increase evident for the financial variables.⁹ The 68% error bands suggest that the null hypothesis of no change can be rejected. Looking at individual countries, this increase in the contribution of common uncertainty appears to be a general phenomenon as far as interest rate and stock return variance is concerned. The change in the contribution is especially larger for the United States, Canada, Germany, France, Italy and Australia. A similar pattern is evident for the change in the contribution of common uncertainty to inflation variance, with the change estimated to be greater than 10% for the United States,

Italy, Spain and Australia. The results for the contribution of common uncertainty to output growth variance are more heterogeneous. While countries such as Canada, France and Australia appear to have experienced a moderate to large increase in the contribution of common uncertainty, the estimated change is small or statistically equal to zero for the United States, Germany and Spain.

The counterpart to the increasing importance of the common component is a decline in the contribution of the country-specific and idiosyncratic components for output growth, inflation and stock returns and a decline in the role of the idiosyncratic uncertainty in the case of the short-term interest rate.

In summary, the variance decomposition exercise suggests three results. First, we find that the role played by 'world' uncertainty in driving the variance of real and nominal variables is non-trivial on average. Second, common uncertainty appears to be, on average, more important for the variance of inflation and stock returns than output growth. Finally, the contribution of common uncertainty is estimated to be higher after the 1990s and the co-movement in volatility of key variables has increased. This phenomenon is especially evident for inflation, interest rate and stock return variance. In the next section, we test if these results are robust to changes in the specification of the benchmark model.

3.3. Sensitivity analysis

In order to test the robustness of these results, we carry out key sensitivity checks. While our model features time-varying variances, the parameters of the observation and transition equations are assumed to be fixed. To investigate the impact of this assumption, we re-estimate the model over the post-1985 period. This period corresponds roughly to a shift in macroeconomic dynamics for OECD countries, with lower volatility and lower inflation signalling the start of the 'Great Moderation'. The resulting variance decomposition is shown in figure 2 in the technical Appendix. The figures show that the key results are preserved. In particular, common uncertainty is estimated to be important in driving the variance of output growth, inflation, interest rates and stock returns with the average contribution increasing after the late 1990s. As a second check, we re-estimate the benchmark model restricting the sample to 2006Q4. This is done to account for the possibility of large parameter changes over the recent financial crisis. Figure 3 in the technical Appendix again shows that the main features of the variance decomposition are preserved in this alternative specification.

4. Explaining the empirical results. A DSGE model

The empirical analysis in the paper suggests that the time-varying volatility of key macroeconomic and financial variables is characterised by a common component across countries and that this

⁸ Discrete values for the interest rate introduces occasional zeros in the residuals leading to numerical problems in the stochastic volatility part of the MCMC algorithm.

⁹ Note that, if the post-1990 average is calculated excluding the period after 2006Q4, the same conclusions are reached. The average percentage change in the contribution of the common component [68% error bands] is 4.6 [2.3, 6.7], 7.0 [4.8, 9.4], 10.3 [7.6, 14.0], and 8.2 [5.0, 11.6] for the variance of output growth, inflation, interest rate and stock returns, respectively.

Table 3

Percentage difference in the contribution to the variance post- and pre-1990. Positive numbers indicate that the contribution was higher after 1990.

	Output growth			Inflation		
	Common	Country	Idio.	Common	Country	Idio.
United States	1.5 [0.7, 2.7]	0:1 [−1.3, 1.7]	−1.5 [−3.4, −0.4]	16:2 [11.0, 20.6]	−7.6 [−12.6, −3.6]	−7.8 [−12.6, −4.0]
United Kingdom	5.0 [2.5, 10.9]	−3.3 [−8.0, −0.6]	−1.6 [−4.6, 0.21]	3.0 [1.6, 5.9]	−3.3 [−6.1, −1.8]	0.2 [0.1, 0.4]
Canada	21.0 [14.8, 28.0]	18.6 [12.3, 24.2]	−39.8 [−44.8, −34.6]	2.9 [−1.0, 6.6]	−6.6 [−10.5, −2.5]	3.5 [0.5, 7.3]
Germany	2.0 [−2.0, 5.8]	−12.2 [−17.6, −7.4]	10.2 [5.5, 16.0]	4.9 [3.0, 8.0]	−5.32 [−8.6, −3.2]	0.5 [0.1, 0.9]
France	16.5 [10.4, 21.5]	−4.6 [−9.8, −1.64]	−10.5 [−16.4, −4.9]	6.2 [2.7, 11.0]	−0.3 [−5.0, 3.7]	−6.4 [−12.2, −0.74]
Spain	−3.5 [−8.0, 0.2]	−2.8 [−10.6, −0.6]	8.24 [3.18, 15.03]	27.8 [23.4, 31.4]	−26.4 [−30.0, −22.03]	−0.9 [−2.4, −0.41]
Italy	17.2 [12.6, 21.8]	−12.1 [−16.9, −8.0]	−4.6 [−10.1, 0.36]	19.2 [14.4, 24.0]	−18.9 [−23.3, −14.0]	−0.3 [−0.8, −0.1]
The Netherlands	13.3 [9.3, 18.1]	7.8 [2.6, 14.0]	−21.6 [−27.3, −16.2]	18.7 [14.8, 22.4]	−18.5 [−22.3, −14.6]	−0.1 [−0.2, −0.05]
Sweden	−2.0 [−4.8, 0.07]	−3.3 [−8.0, 0.62]	5.6 [0.50, 11.54]	−2.3 [−6.2, 0.6]	2.4 [−0.5, 6.3]	−0.1 [−0.2, −0.06]
Japan	4.3 [1.3, 8.0]	−3.6 [−8.0, 0.20]	−0.87 [−5.6, 3.8]	5.2 [2.6, 8.0]	−4.1 [−7.0, −1.4]	−1.0 [−1.8, −0.7]
Australia	9.0 [6.2, 12.6]	−9.3 [−13.0, −6.63]	0.35 [0.21, 0.54]	15.0 [11.5, 19.5]	−14.9 [−19.4, −11.5]	−0.04 [−0.1, 0.01]
Average	8.0 [5.5, 10]	−2.6 [−4.2, −0.12]	−5.0 [−7.0, −3.3]	10.8 [8.5, 13.0]	−9.6 [−11.6, −7.5]	−1.2 [−1.9, −0.5]

Table 4

Percentage difference in the contribution to the variance post- and pre-1990. Positive numbers indicate that the contribution was higher after 1990.

	Interest rate			Stock ret.		
	Common	Country	Idio.	Common	Country	Idio.
United States	11.6 [6.4, 17.7]	5.7 [−0.2, 11.7]	−18.0 [−22.5, −13.2]	12.0 [7.7, 16.5]	−6.7 [−11.6, −2.0]	−4.8 [−10.5, −0.4]
United Kingdom	2.4 [0.8, 5.9]	1.9 [−2.5, 6.9]	−5.0 [−11.3, −0.3]	3.5 [2.0, 5.6]	−2.4 [−4.4, −1.2]	−1.0 [−1.6, −0.6]
Canada	11.1 [4.3, 19.5]	7.2 [3.7, 11.2]	−19.7 [−28.4, −9.7]	9.5 [5.0, 14.5]	0.3 [−0.2, 1.4]	−10.0 [−14.8, −5.5]
Germany	24.4 [16.7, 30.0]	0.3 [0.02, 1.1]	−25.0 [−30.2, −17.4]	9.4 [3.0, 14.3]	−2.1 [−6.1, −0.2]	−6.1 [−11.5, −0.1]
France	18.2 [7.4, 27.4]	0.04 [−0.1, 0.3]	−18.3 [−27.3, −7.6]	28.1 [24.0, 32.7]	−10.6 [−16.0, −5.7]	−17.4 [−24.0, −11.1]
Spain				13:4 [9.0, 18.0]	−19.0 [−25.2, −10.0]	5.5 [−2.7, 12.2]
Italy	11.4 [6.0, 16.8]	−4.2 [−7.2, −1.5]	−6.5 [−12.4, −1.6]	28.7 [23.7, 33.2]	−3.6 [−5.8, −1.6]	−25.0 [−30.1, −19.1]
The Netherlands	13.7 [8.0, 20.1]	−4.8 [−9.8, 0.0]	−8.4 [−15.1, −3.7]	11.3 [5.0, 17.6]	−2.1 [5.0, −0.7]	−8.6 [−15.1, −1.7]
Sweden	13.4 [5.1, 24.3]	3.2 [0.8, 10.9]	−19.5 [−31.1, −8.6]	4.3 [1.2, 12.1]	−11.5 [0.7, 10.1]	−11.5 [−16.9, −6.9]
Japan				5.1 [2.5, 8.2]	−5.7 [−8.7, −2.8]	0.5 [0.2, 0.8]
Australia	26.5 [15.1, 33.8]	−0.1 [−0.3, 0.1]	−26.3 [−33.7, −15.3]	17.6 [14.6, 21.1]	−18.6 [−8.7, −2.8]	0.8 [0.6, 1.2]
Average	14.6 [11.1, 19.0]	1.4 [0.01, 2.6]	−16.2 [−20.0, −12.7]	13.2 [10.3, 16.4]	−6.0 [−7.7, −4.4]	−7.0 [−9.5, −4.9]

co-movement has become more important over time. In this section, we build a DSGE model to provide a theoretical explanation for these results. The proposed New Keynesian model features two-countries with recursive preferences (Epstein and Zin, 1989; Weil, 1989, 1990) long-run risk (Bansal and Yaron, 2004; Rabanal and Rubio-Ramirez, 2015; Rabanal et al., 2011).

The works of Colacito and Croce (2013), Kollmann (2015) and Gourio et al. (2013) have also employed recursive preferences and long-run risks in order to understand the correlation patterns between asset prices and macroeconomic aggregates observed in the data across countries. Similar to Benigno et al. (2011), the model presented here can be viewed as an extension of this previous work along several dimensions: we model the supply side

more comprehensively as a production economy and add nominal frictions.

4.1. Summary of the model

Households in each country form Epstein-Zin preferences, consume, supply labour and invest on a internationally traded riskless bond. Part of the consumption is produced domestically and the rest is imported from the foreign economy. On the supply side, there is a continuum of monopolistically competitive firms that produce an intermediate good by combining labour and (fixed) physical capital. The output of this process is used for the production of the final good,

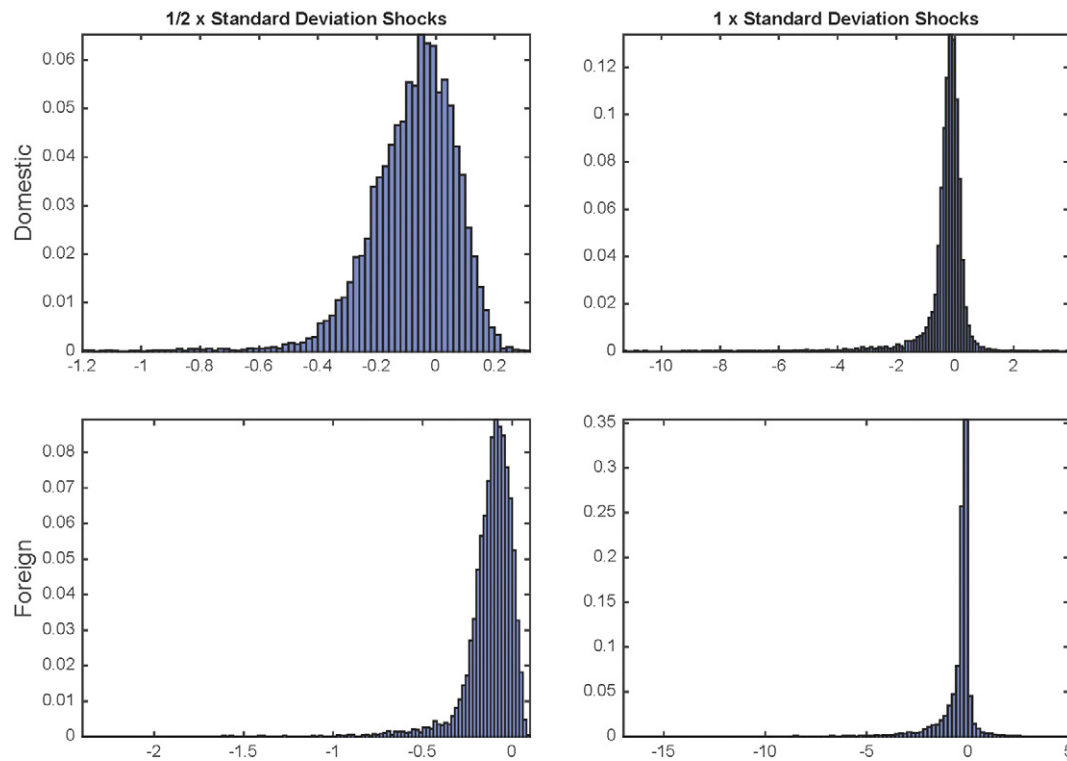


Fig. 4. Starting from the stochastic steady state, the model is simulated from 50,000 periods using only the foreign financial shock. The histogram illustrates the distribution of the domestic and foreign GDP as deviation from their stochastic steady-state.

which can be either consumed domestically or exported, by a perfectly competitive sector. Similarly, the foreign consumption good is imported by a perfectly competitive sector. Prices in the intermediate sector are based on Calvo (1983) schemes, while import prices obey the law of one price. Monetary authorities set policy based on a Taylor type rule (Taylor, 1993). Each economy is disturbed by a non-stationary productivity, a stationary productivity and a financial shock. The two stochastic trends are assumed to be co-integrated to ensure the existence of a balanced path. Furthermore, the financial shock is the one used by Smets and Wouters (2007) – i.e. it decreases the value of the bond exogenously. In short, our theoretical set-up can be viewed as the two-country version of the model developed by Rudebusch and Swanson (2012) and Swanson (2015). Technical details on the model variables and equations can be found in the technical Appendix of the paper. The Appendix also provides details on the calibration of the parameters. The calibration is largely standard and uses the parameter values proposed in previous studies.

4.2. Key implications from the model

4.2.1. Heteroscedasticity

It is interesting to note that this model implies that the volatility of endogenous variables is time-varying without the explicit need for heteroscedastic shocks. As explained by Rudebusch and Swanson (2012) and Swanson (2015), this heteroscedasticity is a consequence of the nonlinearity of the lifetime preference function as well as the fact that this is a production economy. According to the authors, the existence of labour in the model, the fact that shocks do not enter in the model multiplicatively with respect to wealth and the property that agents' wealth consists of both human and physical capital makes the utility kernel non-homothetic. This induces a small degree of conditional heteroscedasticity, which is further enhanced by the high risk aversion.

As discussed by Theodoridis and Mumtaz (2015), the economic intuition behind this is as follows: The non-homotheticity in household's preferences makes agents' responses to economic shocks depend on the current level of the state of the economy. For instance, when the current level of consumption is low (or the marginal utility of consumption is high) then consumption uncertainty is higher (relative to the case where the initial level of consumption/output is high) and this reflects agents' elevated concerns about future shocks. As an adverse shock that lowers output further is going to induce a proportionally larger reduction in utility relative to the case where the initial level of consumption/output was high. This channel is further enhanced by Epstein-Zin preferences as the risk parameter reflects how much agents dislike elevated future uncertainty (Piazzesi and Schneider, 2007; Rudebusch and Swanson, 2012).

Loosely speaking, agents in this economy price adverse shocks more heavily in 'bad times' when compared to 'good times'. This behaviour induces a 'wedge' between the mode of the distribution of the state of the economy and its mean as the latter captures these elevated concerns.

Fig. 4 illustrates this phenomenon. We use the model developed in this study to simulate the data using only the foreign financial shock.¹⁰ The top row of Fig. 4 shows the probability density function of domestic GDP when 1/2 and 1 times the standard deviation foreign financial shocks are used. The second row reports the same information for the foreign GDP. It is apparent that even under 1/2 times standard deviation financial shocks, the probability density functions display a 'downward risk', meaning that the average GDP is below its mode. Furthermore, the asymmetry becomes more pronounced as the size of the shocks increase, indicating that agents' economic behaviour is also a function of the state of the economy.

¹⁰ The technical Appendix presents these simulations using the supply shock. The results are very similar to those obtained using the financial shock.

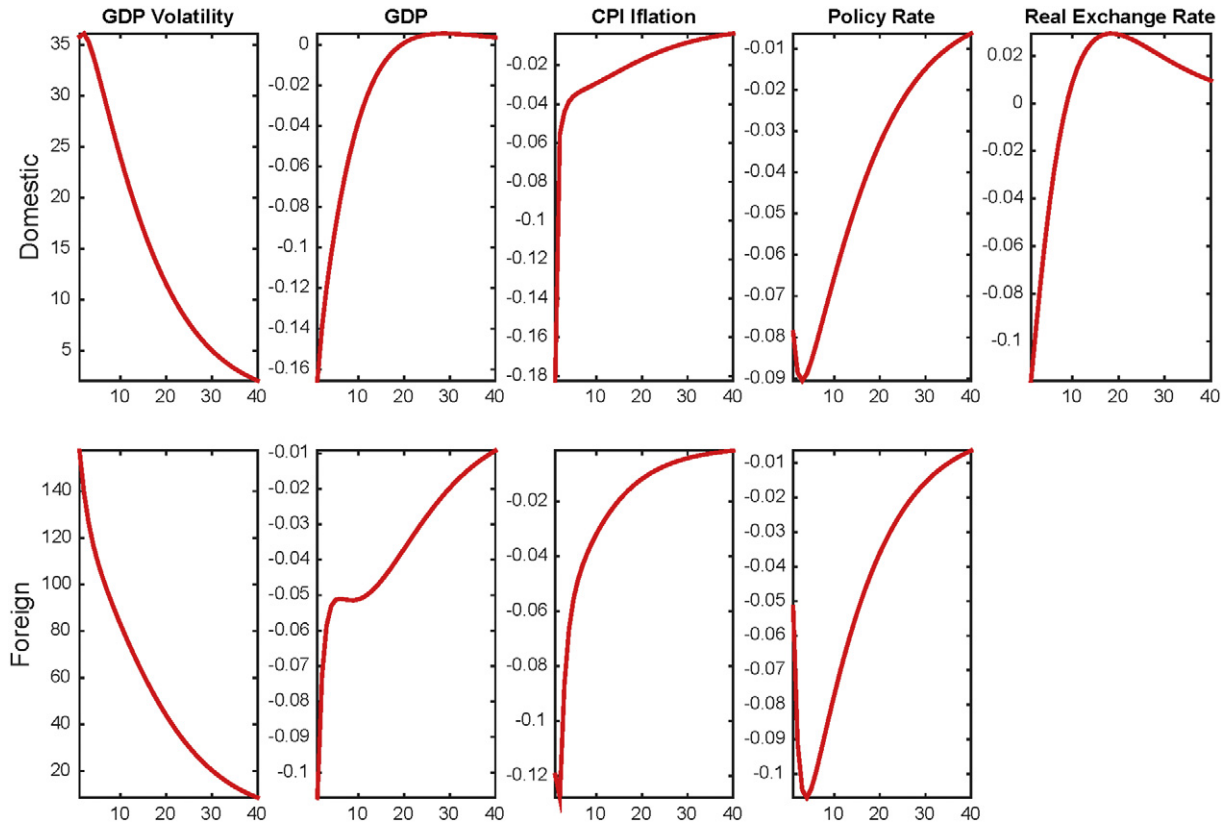


Fig. 5. Response to a financial shock. See Eq. (13) for the definition of GDP volatility. The responses are calculated relative to the stochastic steady state (Juillard and Kamenik, 2005).

Fig. 5 elaborates further on the mechanism that delivers these non-linearities. Following Basu and Bundick (2011) and Swanson (2015), we define the volatility of domestic and foreign GDP as follows:

$$\hat{\sigma}_{Y_t}^2 = 100 \ln \frac{\sigma_{Y_t}^2}{\sigma_Y^2} \quad (13)$$

$$\hat{\sigma}_{Y_t^*}^2 = 100 \ln \frac{\sigma_{Y_t^*}^2}{\sigma_{Y^*}^2}$$

$$\sigma_{Y_t}^2 = E_t(Y_{t+1} - E_t Y_{t+1})^2$$

$$\sigma_{Y_t^*}^2 = E_t(Y_{t+1}^* - E_t Y_{t+1}^*)^2$$

where σ_Y and σ_{Y^*} are the stochastic steady-state values of the domestic and foreign GDP volatility, respectively.¹¹ The financial shock in this study is similar to the household's budget constraint (Smets and Wouters, 2007) bond shock and it can be thought of as a 'haircut' to the financial wealth (the value of the bonds).¹² This shock increases the effective interest rate faced by foreign households and this suppresses demand and inflation. However, in our model there is an additional channel that lowers demand further and this is uncertainty. Foreign agents see that their consumption falls but they are also assigned a non-zero probability to a future shock that is more adverse in terms of its impact on consumption. As they dislike future uncertainty, they reduce consumption further to hedge against future consumption risks. These elevated concerns are

captured by the mean of the state vector ($E_t Y_{t+1}^*$) and this opens up a 'wedge' between Y_{t+1}^* and $E_t Y_{t+1}^*$ and volatility rises.

The fall in the foreign demand and, consequently, domestic exports contributes to the volatility increase in the domestic economy (via a similar mechanism like the one discussed above). Moreover, as explained by Colacito and Croce (2013), domestic agents realise that the difficulties faced by the foreign economy limit their ability to hedge against future adverse domestic shocks via risk sharing. In other words, as the degree of risk sharing decreases, domestic volatility rises (see Fig. 4).

In Fig. 6, we plot the realised volatility of domestic and foreign output, inflation, policy rate and equity returns when all shocks are active. These simulations further confirm that the data generated by the model with homoscedastic shocks display time-varying variance.¹³

4.2.2. Co-movement in volatility

The model also implies cross-country co-movement in the volatility of endogenous variables. First, as the two economies are symmetric, agents in both countries display precautionary behaviour in response to shocks. Furthermore, shocks originating in the domestic economy are transmitted to the foreign one via asset links (UIP condition) or/and good (imported consumption) trade links. These channels: (i) magnify the precautionary behaviour (see Colacito and Croce, 2013, Appendix A) ii) cause higher moments to be

¹³ The volatility is calculated via simulation. The model is simulated for 100,000 periods. The first 50,000 observations are discarded to eliminate the effects of the initial conditions. The remaining 50,000 periods are used to calculate the realised standard deviation of these series based on a 40-quarter rolling window. A very similar picture is obtained when a uni-variate stochastic volatility model is used to estimate the realised volatility of the simulated data.

¹¹ The stochastic steady states are calculated as in Juillard and Kamenik (2005).

¹² Smets and Wouters (2007) view this perturbation as a reduced-form net-worth shock.

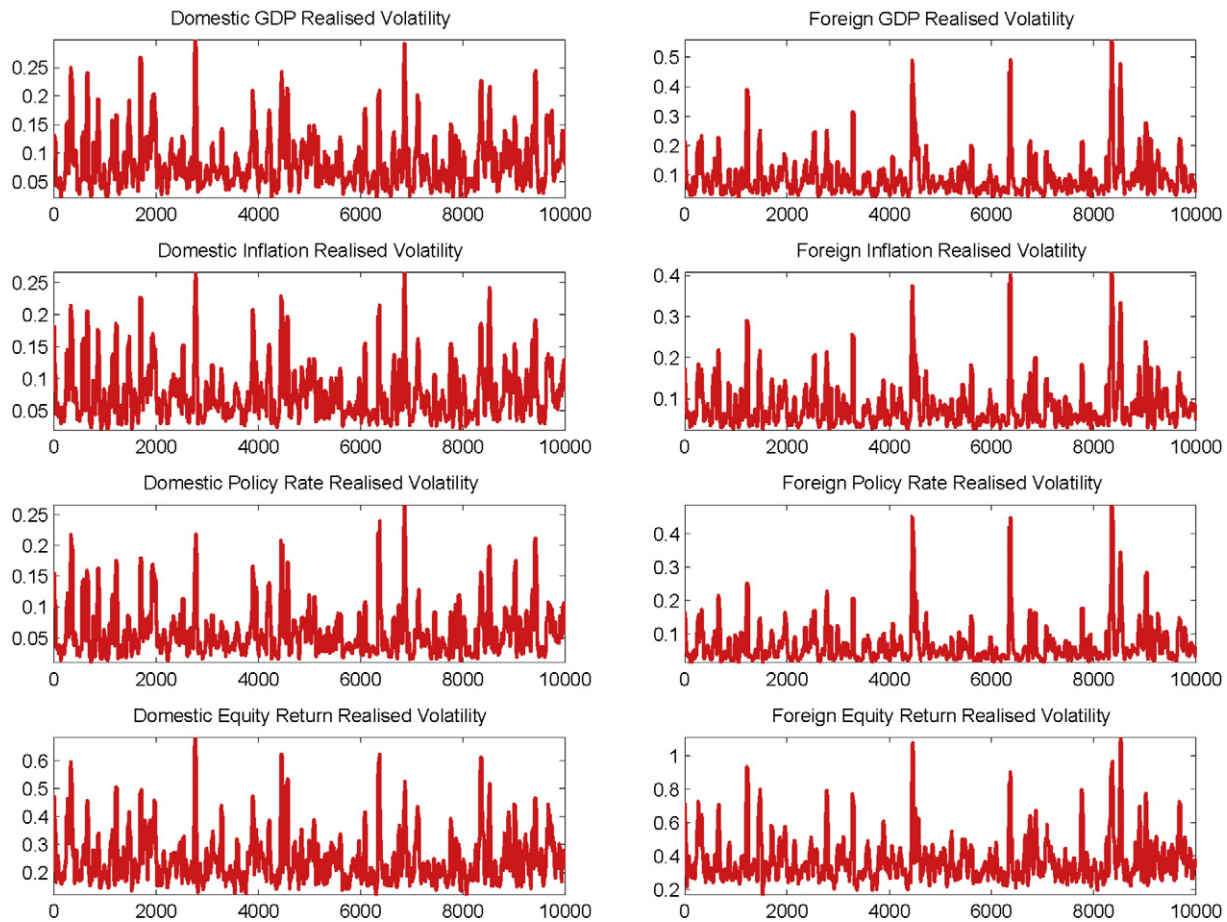


Fig. 6. Realised volatility of key model variables.

correlated across the two economies. For example, consider a situation where a positive shock increases the supply of the domestic good and thus reduces domestic uncertainty. As the home economy marginal utility of consumption drops, home economy agents find it optimal to transfer resources to the foreign country. This transfer of resources has an ameliorating impact on the volatility of foreign variables and thus induces a correlation between the second moments across countries.

In order to demonstrate this feature, we calculate the correlations across countries of the simulated volatilities shown in Fig. 6. The first column of Table 5 displays the results under the benchmark calibration.

The estimated correlations capture a number of features highlighted by the empirical results. First, the estimated correlations are non-negligible and show that the volatility co-moves across countries. Second, the correlation of the volatility of output is lower than the correlation of the variance of inflation and the financial variables. This feature is driven by the precautionary mechanism described earlier, where agents with recursive preferences are keen to give up expected future consumption in order to ensure a less volatile future consumption profile and this leads to a transfer of consumption across countries. For this transfer to occur, prices and especially the exchange rate need to adjust in order for the markets to clear. Since domestic prices are staggered (Calvo contracts) most of the variability of the CPI inflation is driven by import prices and, as the law of one price holds for importing firms, by the exchange rate variability. Due to the symmetry of the model, exchange rate uncertainty has very similar effects on CPI inflation in both countries

and this induces the higher moments of CPI inflation across countries to co-move strongly. Strong cross-country CPI inflation volatility co-movement implies that the interest rate variance across countries is correlated as monetary authorities stabilise inflation by adjusting the policy rate. In contrast, the equity price volatility correlation is driven by the stochastic discount factor. Equity prices are functions of the stochastic discount factor. As the UIP condition attempts to equalise the stochastic discount factor across countries (risk sharing), this results in a closer co-movement of volatility of this variable.

The second column of Table 5 presents second moment correlations estimated assuming a stronger home bias. A comparison with the benchmark correlations makes it clear that as the degree of home bias falls, the second moment correlations rise dramatically. As the economy becomes more open, agents are more willing to hedge against future risks about expected utility by transferring resources abroad. Therefore, globalisation and an increase in trade provides one explanation for the empirical result that the role of the common uncertainty component has increased over time and consequently, the volatility of output, inflation and financial variables has become more correlated. Fig. 7 shows that the ratio of total trade to GDP, a measure of trade intensity, rose sharply in the countries in our panel after 1990. This date broadly coincides with the increase in the contribution of common uncertainty evident in Tables 3 and 4 and again points to the role of globalisation in driving the second moment co-movement estimated by the empirical model.

Of course, a number of other structural changes also took place in the OECD and globalisation is not the only possible explanation for

Table 5

Correlation amongst the time-varying variance of output (σ_t^Y), inflation (σ_t^π), interest rates (σ_t^R) and equity returns ($\sigma_t^{Q^E}$) across countries. The superscript * denotes the foreign country.

Moments	Benchmark	Strong home bias	Smaller response to inflation	Lower price/wage rigidity
$\text{Corr}(\sigma_t^Y, \sigma_t^{Y*})$	0.60	0.28	0.63	0.80
$\text{Corr}(\sigma_t^\pi, \sigma_t^{\pi*})$	0.77	0.36	0.69	0.94
$\text{Corr}(\sigma_t^R, \sigma_t^{R*})$	0.68	0.35	0.64	0.86
$\text{Corr}(\sigma_t^{Q^E}, \sigma_t^{Q^{E*}})$	0.83	0.46	0.75	0.77

the increased importance of common uncertainty. The DSGE model allows us to consider the plausibility of alternative explanations for this result. First, numerous studies have shown that after the mid-1980s, a number of countries in our panel changed their practice of monetary policy and adopted a more anti-inflationary stance (see for example Clarida et al., 1998). In order to consider the role of systematic policy, we re-calculate the cross-country volatility correlation under the assumption that the coefficient on inflation in the Taylor rule in both countries falls from 1.5 to 1.01. As monetary authorities do not target inflation aggressively, price dispersion rises, magnifying the resource costs associated with sticky prices. This increases the unconditional variance of real and nominal variables in each country but, as shown in the third column of Table 5, does not change the second moment correlations relative to the benchmark case. This result suggests that as long as preferences of policy makers regarding inflation are symmetric across countries, any change on the weight placed on inflation does not trigger a change in the way that agents try to ensure against future utility risks. Given that anti-inflationary monetary policy was in place across our panel of countries after the mid-1990s, a change in the practice of monetary policy is less likely to be an explanation for the increasing role of common uncertainty. Fernandez-Villaverde and Rubio-Ramirez (2008) and Hofmann et al. (2012) present evidence showing that the degree of price and wage nominal rigidities increased during the Great Moderation. The fourth column of Table 5 presents the estimated second moment correlations assuming a lower degree of nominal rigidities than the benchmark case. A comparison of these

results with the benchmark correlations suggests that a rise in nominal rigidities is associated with decreased co-movement in some of the volatilities thus casting doubt on this structural shift as a factor behind the increased role played by the common uncertainty component.

5. Conclusions

This paper uses a factor model with stochastic volatility to decompose the time-varying volatility of output, inflation, interest rates and stock returns into contributions from country-specific uncertainty and uncertainty common to all countries. We find that the common component plays an important role in driving the time-varying volatility implying that the second moments are correlated across countries. The empirical results suggest that the role of the common component has increased after the mid-1990s and the second moments of real and financial variables have become more correlated in the recent past. The correlations are estimated to be larger for nominal and financial variables.

In an attempt to provide an economic explanation for these results, we build a two-country DSGE model featuring Epstein-Zin preferences. One of the key implications of this preference set-up is that the volatility of endogenous states in the model is time-varying. Moreover, agents are keen to give up expected future consumption in order to ensure a less volatile future consumption profile and this leads to a transfer of consumption across countries.

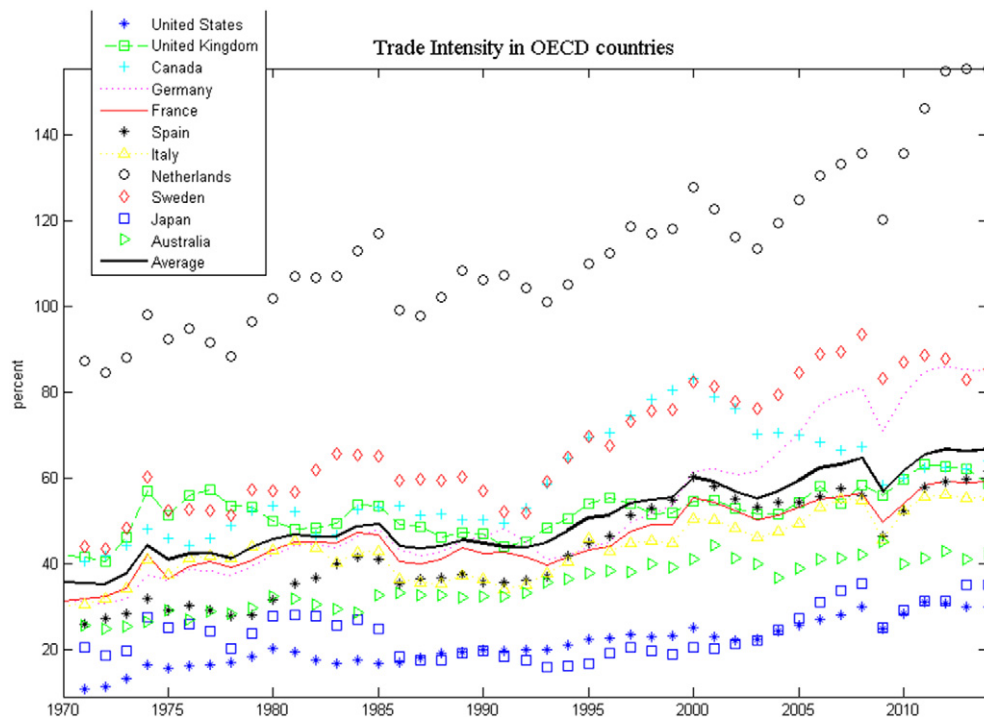


Fig. 7. Ratio of total trade to GDP.
Source: Source: World Bank.

These transfers and risk-sharing result in a close co-movement in second moments across countries. Simulations from the model suggest that an increase in trade openness leads to a closer movement in volatilities and thus provides one explanation for the increasing importance of the common uncertainty factor suggested by the empirical model.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jinteco.2017.01.007>.

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